Singapore & Asian Schools Math Olympiad

Secondary 2 (Grade 8) – GEP Practice

2019 & 2020

Contest Problems with Full Solutions

Authors:

Henry Ong, BSc, MBA, CMA
Merlan Nagidulin, BSc

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About the Authors:

Henry Ong is the Founder of SASMO and President of Singapore International Mastery Contest Centre (SIMCC). He conducts Math Olympiad classes, not only in Singapore but also around the 20 countries that have joined SASMO. He now travels round the world giving out awards and scholarships to students and teachers. He is now promoting new contests, International Scholastic Trust Teachers' Institute (ISTTI) and International Junior Honor Society (IJHS).

In February 2020, Henry was appointed Co-Chair of the Council Administration of Cambodian Mathematical Society (CA_CMS). Henry will share and use his great wealth of experience, expertise, and international networks to contribute effectively to: (1) promoting STEM: Science, Technology, Engineering, and Mathematics; (2) organizing and undertaking multilateral partnership programs; and (3) strengthening and improving the public-private partnership for development of STEM, research and development, and research and innovation as well as teacher professional development in Cambodia with relevant stakeholders in close collaboration and consultation with relevant leadership and management of Ministry of Education, Youth and Sport, especially Cambodian Mathematical Society (CMS), New Generation Pedagogical Research Center (NGPRC), National Institute of Education(NIE) and Teacher Training Department (TTD).

Merlan Nagidulin is the Academic Director for Math and Science Olympiad and General Manager of SIMCC. Merlan graduated with a Bachelor of Science, Honours in Mathematical Sciences from Nanyang Technological University. Merlan won 2 gold medals in Kazakhstan National Math Olympiad and was awarded a bronze medal in the most prestigious International Math Olympiad (IMO Slovenia 2006).

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Noble Education Pte Ltd – 55 Lor L Telok Kurau, 03-69, Bright Centre, Singapore 425500
SIMCC – 199A Thomson Road, Goldhill Centre, Singapore 307636 Website: www.simcc.org

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International Scholastic Trust President's Message

Dear SIMCC contestants, parents, teachers, and partners,

I hope all of you are safe and healthy. We have all gone through so much within the past year, and we should rejoice - we are all alive and well. Despite all the groom and predictions of the worst recession to hit us very soon, we at SIMCC, are expanding globally – adding another 14 country partners to a total of 35 countries and territories within the last 10 months, and setting up sales offices in Cambodia and India, in addition to taking over the ICAS competitions and staff from the University of New South Wales Global Office based in Hong Kong and Macau.

Two years, Sophie Koh, our Senior Director of IT, Events and Operations left Singapore Polytechnic as a Senior Lecturer of IT to join us. She has almost digitally transformed most of our operations and contest systems. This will provide valuable big data in formative and summative assessments for teachers to measure their students and inform their teaching. Parents and students will also receive Detailed Performance Reports (DPR) in our Member Development Portal (MDP) to improve their Academic Skills. Teachers and their schools will be able to download Students' Analytical Performance Reports (SAPR). SIMCC together with our partners will be conducting more Webinars to prepare teachers and parents to use our contests and data to arm their wards with Skills and Recognition for SUCCESS.

From February 16 to April 30, 2021, we are offering the 1st Singapore Math Global Assessment for grades 1 to 11 students. We want to open this up to every country, especially since Singapore Math textbooks are used in schools from over 70 countries. We want to share our rich Singapore Math heritage that has propelled Singapore students to the top of PISA rankings since the 1990s. We also have excellent teachers, Learning Management System (LMS), Math textbooks, assessment books, as well as teacher training to help improve math education worldwide. Reach out to us for help, if you want your students to become Mathematics Heroes!

The global winners of SINGA Math Assessments are invited for the 1st SINGA Math Global Finals, in their home countries on September 11, 2021. The prize for the Overall Champion of grades 1 to 10/11 is a S$600 competition package to compete in the STEAM AHEAD International Junior Mathematics Olympiad (IJMO) to be held in Bali, Indonesia from December 3 to 7, 2021. The prize for Overall Runner-up is S$250 and 2nd Runner-up is S$150, offering a total prize award of S$10,000. Winners of Gold (10%) and Silver (15%) are invited to compete in STEAM AHEAD IJMO 2021.

From 2021, we will be offering most of our contests to grade 12 students – SASMO, DrCT, SIMOC, and STEAM AHEAD. This will pave the way for Grades 12 SIMCC students to apply for SIMCC and STS's US$270,000 endowed 2 scholarships for 4-year undergraduate studies at Southern Illinois University (SIU), Carbondale, Illinois, USA in STEM related fields. SIU has also given SIMCC over 30 SIU International Student Tuition Grants to award to top SIMCC students to pursue their university education. SIMCC will setup our own network for SIMCC scholars at SIU. This will help us expand SIMCC contests in North America and generate revenue for us to fund more scholarships.

Best Regards,
Henry Ong, President

Competition and Prizes

SASMO is devoted and dedicated to bringing a love for Mathematics to students. Unlike most Math Olympiad Competitions, SASMO caters not only to students in the top 5% but to the top 40% instead. It aims to arouse students' interest and enthusiasm for mathematical problem solving, develop mathematical intuition, reasoning and logical thinking, as well as creative and critical thinking. In addition, this can help improve the students' math grades because they can apply problem-solving strategies learnt during the training to their daily school mathematics.

History:

Created in 2006, SASMO is one of the largest Math Olympiads in the Asian region. We have expanded the competition to provide an International platform for students from Primary 2 to Secondary 4, with differentiated contest papers for every level. SASMO awards medals and certificates to the top 40% of participants.

Calculators are not permitted

When a problem introduces a more advanced concept, all necessary definitions are included.

Awards:

Each participant receives a Certificate of Participation or an award certificate for winners below. Each of the top 8%, 12% and 20% of all participants receives a Gold, Silver or Bronze medal and certificate respectively. Each student who achieves a Perfect Score of 85 points receives a Perfect Score certificate, Gold medal and $100.

Introduction

For Students Taking the Math Olympiad Challenge

Congratulations. You have embarked on a journey of scholarship. Competitions like SASMO open many doors for you. Firstly, you learn new and interesting approaches to problem-solving and new topics. Next, you will meet talented students from other schools as you attend trainings and competitions. You build your endurance to "puzzle" out challenging problems and build your reputation as a problem solver. Finally, you will be exposed to various international competitions and scholarship opportunities. Here in Singapore, you increase your chances of getting into a top school via Direct School Admissions (DSA) and entry into the Gifted Education Programme as you compete regularly in high level competitions.

This book is written for the participants in the Singapore and Asian Schools Math Olympiads (SASMO). It helps students to prepare well for the contest and develop higher-order thinking. All problems are designed to help students develop the ability to think mathematically, rather than to teach more advanced or unusual topics. The fun is in how you can see patterns and ways of solving each problem in non-technical ways even though you have not learnt the topic yet!

In addition to the contest problems, the reader is provided with a list of familiar mathematical terms, as well as a review of some of the topics that are likely to be tested in the Olympiad. The book also contains some solved examples to provide different problem-solving techniques, and to familiarize the participant with different types of Olympiad questions. It is advised that the reader spends appropriate time studying these questions and solutions, as they will assist in tackling actual Olympiad problems.

How to Use This Book: Practice daily for 15 minutes per hour instead of 4 hours of learning once a month. Your mind needs to absorb each new thought, and constant practice allows frequent review of previously learned concepts and skills. Together, you can remember many new problem-solving approaches. Try to spend 10 or 15 minutes daily doing two or three problems. This approach should help you minimize the time needed to develop the ability to think mathematically.

Whether you solve a problem quickly or if you are confused, it is worth studying the solutions in this book, because often they offer unexpected insights that can help you understand the problem more fully. After you have invested time – trying to solve each problem any way you can, reviewing our solutions is very effective. Many of the problems in this book can be solved in more than one way. There is always a single answer, but there can be many paths to that answer. Once you solve a problem, go back and see if you can solve it by another method. Then check our solutions to see if any of them differ from yours.

Enjoy working on these challenges and you will soon be in a different league from your peers who have not taken any international competition. We look forward to inviting you if you are a bronze, silver, gold or perfect score medallist for further training.

Problem Solving Procedure

You may go through several phases when solving a problem such as trying to understand the problem, working on a specific approach (planning and attempting), getting stuck and trying to get unstuck, critically examining solutions or communicating. The work may involve going back and forth between these different phases of problem solving.

In solving any problem, it helps to have a working procedure. You might want to consider this four-step procedure: Understand, Plan, Try It, and Look Back.

Understand

Before you can solve a problem, you must first understand it. Read and re-read the problem carefully to find all the clues and determine what the question is asking you to find. What is the unknown? What is the data? What is the condition?

Plan

Once you understand the question and the clues, it's time to use your previous experience with similar problems to look for strategies and tools to answer the question. Do you know a related problem? Look at the unknown! And try to think of a familiar problem having the same or a similar unknown?

Try It

After deciding on a plan, you should try it and see what answer you come up with. Can you see clearly that the steps are correct?
But can you also prove that the steps are correct?

Are you feeling stuck?

Many different approaches can be tried to get unstuck. One approach is to try working a simpler version of the problem, and use the solution to the problem to get insights that are useful in solving the original problem. In the next chapter, we will show some common solving approaches.

If you are discouraged after a few failed attempts, read this quote from the famous scientist, Thomas Edison. An assistant asked, "Why are you wasting your time and money? We have had failure after failure, almost a thousand of them. Why do you continue to pursue this impossible task?" Edison said, "We haven't had a thousand failures, we've just discovered a thousand ways to not invent the electric bulb."

Look Back

Once you've tried a problem and found an answer, go back to the problem and see if you've really answered the question. Sometimes it's easy to overlook something. If you missed something check your plan and try the problem again.
Can you check the result?
Can you check the argument?
Can you derive the result differently?
Can you see it at a glance?

Problem Solving Strategies

1. Change the representation

Using a wrong representation may make a problem impossible to solve.
Strategies of changing representation include drawing a picture and acting it out.

DRAW A PICTURE: By drawing a picture, and visualizing the problem information using it, you will have a clearer understanding of the problem and it will help you to come up an approach to solve the problem that you might not be able to see otherwise.

ACT IT OUT: We are better at thinking in terms of concrete objects and situations than in terms of abstract concepts. If we can act out the situation described in a word problem, we are able to understand the problem better and we may be able to come up with a solution. To do this, we need to use real materials that are easily available to us. Examples can be pencils, coins and other objects we have in the classroom.

2. Make an Organized List or a Table

ORGANIZED LIST: Making an organized list allows you to clearly examine data. It can help you in ensuring that you are looking at all the relevant information. It will also allow you to see patterns in the data easily and to come to correct conclusions.

MAKE A TABLE: Making a table allows you to clearly examine data. It can help you in ensuring that you are looking at all the relevant information. It will also allow you to see patterns in the data easily and to come to correct conclusions.

3. Create a Simpler Problem

Sometimes we are not able to solve the problem as it is stated, but we are able to solve a similar problem that is similar in some way. For example, the simpler problem may use simpler numbers. Once we solve one or more simpler problems, we may understand the approach that can be used to solve the problems of similar type and may be able to solve the problem that has been given to us.

4. Use Logical Reasoning

Logical reasoning is useful in mathematics problems in various ways. It can be used to eliminate incorrect choices. It can also sometimes be used to conclude the answer directly.

5. Guess and Check

"Guess and Check" strategy can be used on many problems. If the number of possible answers is small, one can use this strategy to come up with the answer very quickly. In some other cases where the number of possible answers is not small, one may still be able to make intelligent guesses and come up with the answer.

6. Working Backwards

Sometimes, it is easier to start with information at the end of the problem and work backwards to the beginning of the problem than the other way around.

SASMO 2019 Secondary 2 (Grade 8) Contest Questions

Section A (Correct answer – 2 points| No answer – 0 points| Incorrect answer – minus 1 point)

Question 1

In Mathematics, the product of the first n positive integers is written as $n! = n \times (n-1) \times \ldots 1$. For example, $2! = 2 \times 1$ and $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. Evaluate:

$$9 + (8 + 7) \times (6! \div 5 - 4 - 3 - 2 - 1)$$

A. 955
B. 2019
C. 2034
D. 3216
E. None of the above

Question 2

Which of the following is correct?

A. $(a+b)^3 = a^3 + b^3$
B. $-10^2 = 100$
C. $\sqrt{x^2 + 9y^2} = x + 3y$
D. $\frac{x}{x+y} = \frac{x}{x} + \frac{x}{y}$
E. None of the above

Question 3

The highest common factor (HCF) of two positive integers is 3. If the sum of the two integers is 21, which could NOT be their least common multiple (LCM)?

A. 18
B. 30
C. 36
D. 7
E. None of the above

Question 4

If the five-digit number 5D68D is divisible by 36, find the value of D.

A. 0
B. 2
C. 4
D. 8
E. None of the above

Question 5

If $X = 0.0000125$ and $Y = 0.00008$, which of the following has the greatest value?

A. $X + Y$
B. $X - Y$
C. $X \times Y$
D. $X \div Y$
E. None of the above

Question 6

The price of an ice cream cone in a shop has increased by 20%. How many ice cream cones can now be bought using the amount of money that used to buy the same 60 ice cream cones?

A. 50
B. 48
C. 40
D. 56
E. None of the above

Question 7

The following calculations lead to the conclusion that $\pi > 2019$.

Let $x = \pi\ (3.14159\ldots)$, $y = 3$ and $z = 2019$ (1)

$-x \leq -y$ and $z - y > 0$ (2)

$x \geq y$ and $z > y$ (3)

Then $x \geq z$ (4)

Therefore, $\pi > 2019$ (5)

Which of the following transition steps is the first that contains an error?

A. (1) to (2)
B. (2) to (3)
C. (3) to (4)
D. (4) to (5)
E. None of the above

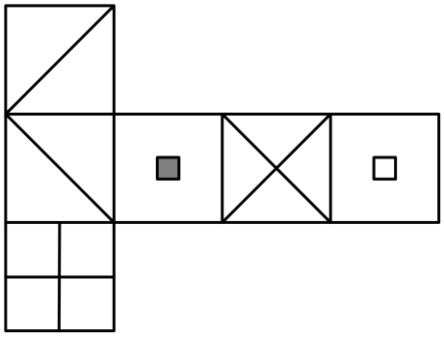
Question 8

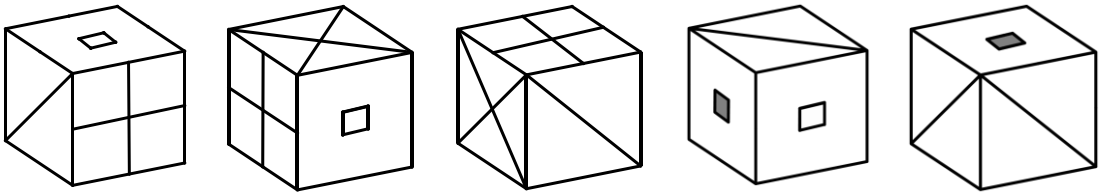
Consider all the integers from 1 to 200. Which of the following is NOT true about them?

A. The sum of all even numbers is greater than the sum of all odd numbers.
B. Writing all the digits, the digit 0 is the least occurring digit.
C. The average of the even numbers is greater than the average of the odd numbers.
D. (Sum of all even numbers) – (Sum of all odd numbers) = 200
E. The average of the 200 numbers is not a positive integer.

Question 9

Which cube below will be formed when the net on the right is folded?





A B C D E

Question 10

How many consecutive zeros are there in the end of $2^{11} \times 3^7 \times 5^5 \times 7^3 \times 11^2$?

A. 11
B. 16
C. 6
D. 5
E. None of the above

Question 11

The sum of the speeds of two cyclists is 34 km/h. If the first cyclist cycles for 3 hours and the second cyclist cycles for 7 hours, then the total distance they travel will be 150 km. Assuming both cyclists cycle at a constant rate, what is the difference between their speeds?

A. 12 km/h
B. 11 km/h
C. 10 km/h
D. 9 km/h
E. 8 km/h

Question 12

Find the value of

$$\left[\sqrt{1+2+3+\cdots+100}\right],$$

where $[x]$ is the greatest integer less than or equal to x. For example, $[13.54] = 13$.

A. 69
B. 70
C. 71
D. 72
E. None of the above

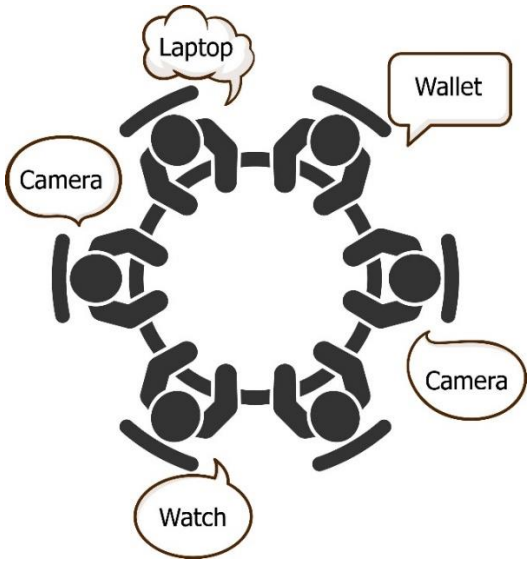
Question 13

Which of the following fractions is the largest?

A. $\frac{60}{119}$
B. $\frac{64}{127}$
C. $\frac{62}{123}$
D. $\frac{63}{125}$
E. $\frac{61}{121}$

Question 14

A leader asked his 6 followers what is inside the gift box he saw. These 6 followers are sitting around a table. Two of them are truth-tellers and they are sitting next to each other. Two followers are liars and they are sitting next to each other as well. Each of the remaining two followers is either a truth-teller or liar, and they are not sitting next to each other. The responses of the followers are shown in the picture below.

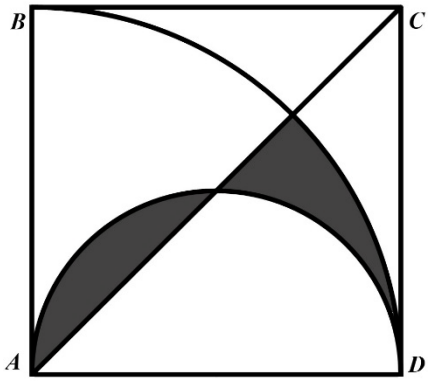


What is inside the gift box?

A. Camera
B. Laptop
C. Wallet
D. Watch
E. Insufficient data

Question 15

In the diagram, $ABCD$ is a square with $AB = 20$ cm, AD is a diameter of the semicircle, and AB is a radius of the quadrant. Find the area (in cm^2) of the shaded region.



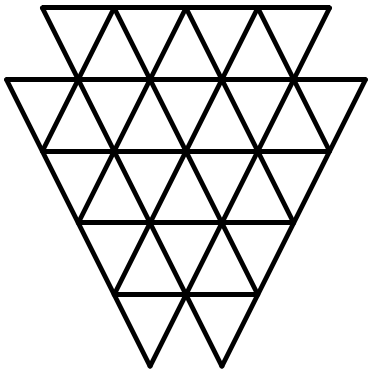
A. 75π cm^2
B. 50π cm^2
C. $(75\pi - 100)$ cm^2
D. $(50\pi - 100)$ cm^2
E. None of the above

Section B (Correct answer – 4 points| Incorrect or No answer – 0 points)

When an answer is a 1-digit number, shade "0" for the tens, hundreds and thousands place.
Example: if the answer is 7, then shade 0007
When an answer is a 2-digit number, shade "0" for the hundreds and thousands place.
Example: if the answer is 23, then shade 0023
When an answer is a 3-digit number, shade "0" for the thousands place.
Example: if the answer is 785, then shade 0785
When an answer is a 4-digit number, shade as it is.
Example: if the answer is 4196, then shade 4196

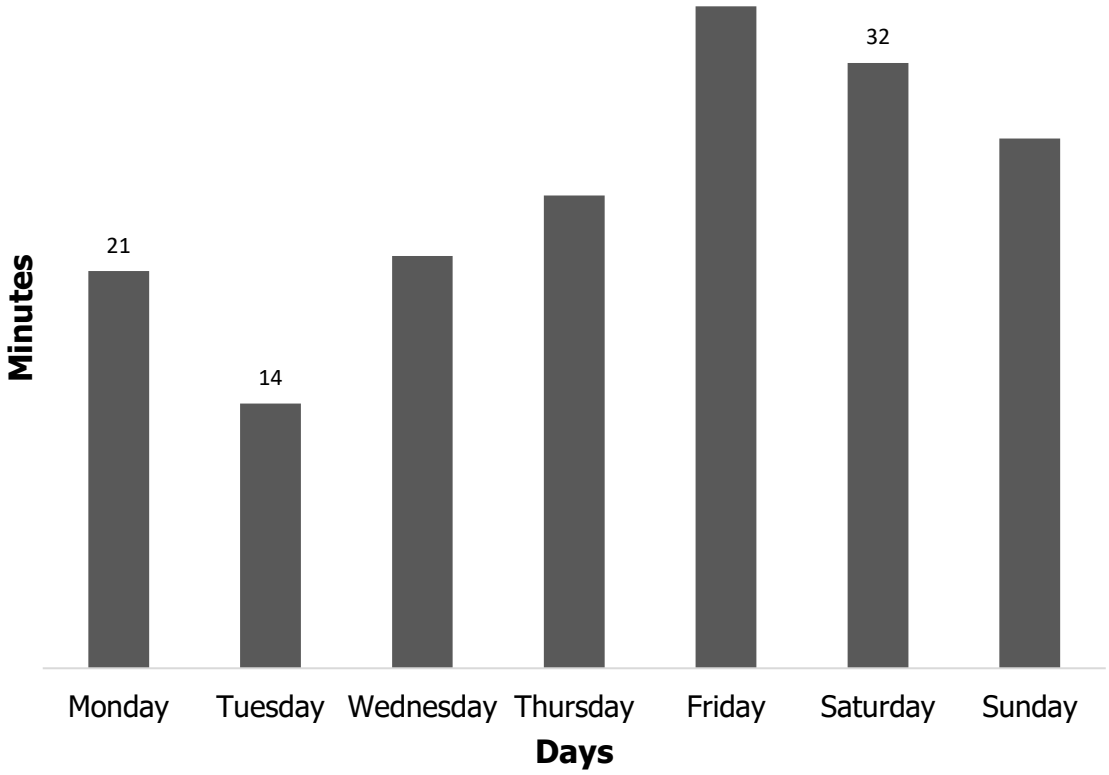
Question 16

How many triangles are there in the diagram?



Question 17

The bar chart below shows the time Lou spent in the gym last week. The average time of the first 3 days was 72% of the average time of the last 4 days. Given the total time spent in the gym last week was 183 minutes, how long, in minutes, did he spend time in the gym last week Wednesday? (Round off your answer to the nearest integer)



Question 18

What is the remainder when $2018! + 13$ is divided by 2019?

Question 19

The operator $\otimes$ acts on two numbers to give the following outcomes:

$$2 \otimes 6 = 44$$

$$4 \otimes 10 = 87$$

$$8 \otimes 18 = 1613$$

$$16 \otimes 22 = 2019$$

Find the value of $7 \otimes 9$.

Question 20

In a camp, 134 students received a burger and 62 received an ice cream. Fifteen students received burger and ice cream but not fries, and 31 students received all the three types of food. Given there are 200 students in this camp, what is the largest possible number of students who didn't get any of the three types of food?

Question 21

The number X is a composite number with all different digits, and the product of its digits is 360. What is the greatest possible value of $\frac{X}{10}$?

(Round off your answer to the nearest integer.)

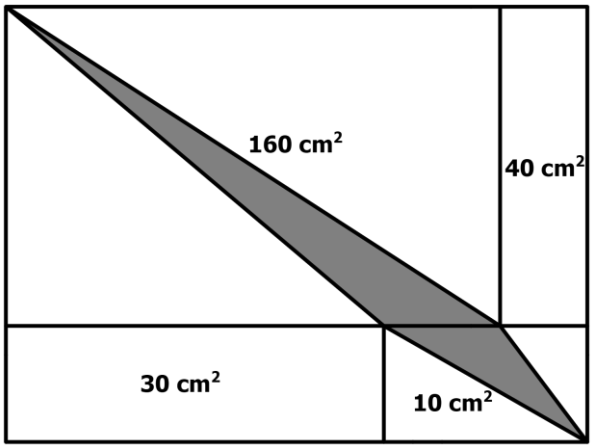
Question 22

Evaluate

$$15120 \times \left(\frac{2}{1 \times 2 \times 3} + \frac{3}{1 \times 2 \times 3 \times 4} + \cdots + \frac{6}{1 \times 2 \times 3 \times \ldots \times 7} \right)$$

Question 23

The large rectangle below is made up of 4 rectangles with their areas indicated in the diagram below. What is the area, in cm^2, of the shaded region?



Question 24

In the following cryptarithm, all the different letters stand for different digits. What is the value of the sum O + D + A + G?

$$\begin{array}{cccccc} & G & O & O & D \\ + & & D & O & G \\ \hline & D & G & D & A \\ \hline \end{array}$$

Question 25

In the equation below, a, b and c are digits, $\overline{aa}$ and $\overline{bb}$ are two-digits numbers, and $\overline{cccc}$ is a four-digit number.

$$\overline{aa}^2 + \overline{bb} = \overline{cccc}$$

What is the largest possible value of $a + b + c$?

END OF PAPER

SASMO 2020 Secondary 2 (Grade 8) Contest Questions

Section A (Correct answer – 2 points| No answer – 0 points| Incorrect answer – minus 1 point)

Question 1

Find the value of the following.

$$2020 \times 20 - 20 \times 20 + 2020$$

A. 2020
B. 42020
C. 40000
D. 40202
E. None of the above

Question 2

Simplify $3^1 \times 3^3 \times 3^5 \times 3^7 \times 3^9 \times 3^{11} \times 3^{13} \times 3^{15} \times 3^{17} \times \left(\frac{1}{3}\right)^{75}$.

A. 27
B. 81
C. 243
D. 729
E. None of the above

Question 3

Which of the following is equal to $ax - 2by + 3c$ if $x = 3a - 4b^2$, $y = 4b - 3c^2$, and $z = -2bc$?

A. $3a^2 - 4ab^2 - 8b^2$
B. $3a^2 - 4ab^2 - 8b^2 + 12bc^2$
C. $3a^2 - 4ab^2 - 8b^2 - 12bc^2$
D. $3a^2 - 4ab^2 + 8b^2 + 12bc^2$
E. None of the above

Question 4

Find the value of N in the equation below.

$$\frac{1+2+3+\cdots+120}{110}=\frac{2+4+6+N}{2}$$

A. 232
B. 242
C. 252
D. 262
E. None of the above

Question 5

A large rectangle in the figure below is divided into nine smaller rectangles. The areas of five rectangles have been given. What is the sum of the areas of the remaining four rectangles?

	1	
$\frac{1}{6}$		$\frac{1}{2}$
$\frac{1}{3}$	$\frac{2}{3}$	

A. $\frac{10}{3}$
B. 2
C. 3
D. $\frac{5}{2}$
E. None of the above

Question 6

Denote $n\Delta m = (n-1)(n-2)\cdots(n-m)$. For example, $5\Delta 3 = 4 \times 3 \times 2$. If $8\Delta x = 840$, what is the value of $x\Delta(x-3)$?

A. 6
B. 12
C. 24
D. 36
E. None of the above

Question 7

The value of $\frac{2022^3-2\times2022^2-2020}{2021^3+2\times2021^2-2023}$ can be written as $\frac{a}{b}$ where $HCF(a,b) = 1$. What is the value of $a+b$?

A. 4039
B. 4041
C. 4043
D. 4045
E. None of the above

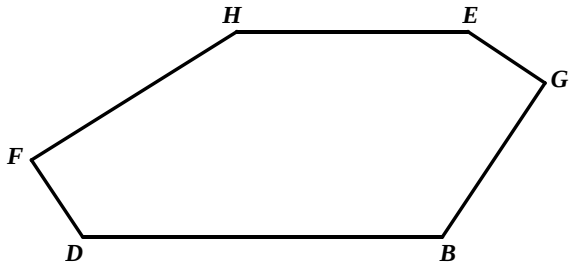
Question 8

If $x < -4$, simplify the expression $\left|1 - \sqrt{(x+3)^2}\right|$.

A. $x + 2$
B. $-x - 2$
C. $x + 4$
D. $-x - 4$
E. None of the above

Question 9

In the figure below, HE is parallel to DB, GE is perpendicular to BG, and DF is also perpendicular to HF. If $\angle H = 138°$, and the measurement of $\angle B$ is 4 degrees more than the measurement of $\angle D$, find the measurement of $\angle E$.



A. $134°$
B. $44°$
C. $46°$
D. $138°$
E. None of the above

Question 10

In the list below, how many fractions are in its simplest form?

$$\frac{1}{2020}, \frac{2}{2020}, \frac{3}{2020}, \ldots, \frac{199}{2020}, \frac{200}{2020}$$

A. 121
B. 59
C. 79
D. 144
E. None of the above

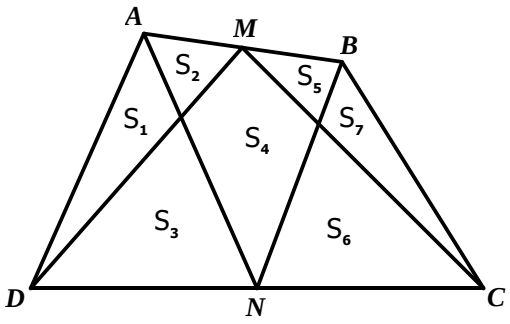
Question 11

Boris wants to write all five-digit numbers using all the digits 1, 2, 3, 4 and 5. The numbers are such that the even digits are in increasing order from left to right, for example 32145 and 25341. How many numbers can Boris write?

A. 45
B. 52
C. 60
D. 64
E. None of the above

Question 12

In quadrilateral $ABCD$, M and N are midpoints of sides AB and DC, respectively. The segments AN, BN, DM and CM divide the quadrilateral into 7 parts $S_1, S_2, S_3, S_4, S_5, S_6$ and S_7 of different areas. Which of the following relationships involving the areas is true?



A. $S_2 + S_4 = S_6$
B. $S_1 + S_7 = S_4$
C. $S_2 + S_3 = S_4$
D. $S_1 + S_6 = S_4$
E. None of the above

Question 13

If today is Tuesday, what day of the week will it be 2^{2020} days from today?

A. Wednesday
B. Thursday
C. Friday
D. Saturday
E. None of the above

Question 14

Ellie, Violet, Audrey, Tom and Chris are Secondary 1 or 3 students. They study in either Greenwich Secondary School or Westwood Secondary School. It is also given that:

- Tom and Chris are from different schools.
- Ellie and Audrey go to the same school.
- Three students go to Greenwich Secondary School and the other two are from Westwood Secondary School.
- Violet and Chris are from the same grade.
- Audrey and Tom study at different levels.
- Three students study in Secondary 1 and the other two students in Secondary 3.

If one of them is Secondary 3 student from Westwood Secondary School, who is that person?

A. Ellie
B. Violet
C. Audrey
D. Tom
E. Chris

Question 15

In Mathematics, the product of the first n positive integers is written as $n! = n \times (n-1) \times \ldots 1$. For example, $2! = 2 \times 1$ and $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. How many integers from 1 to 40 cannot be the number of consecutive digit "0"s at the end of $n!$, for some integer n?

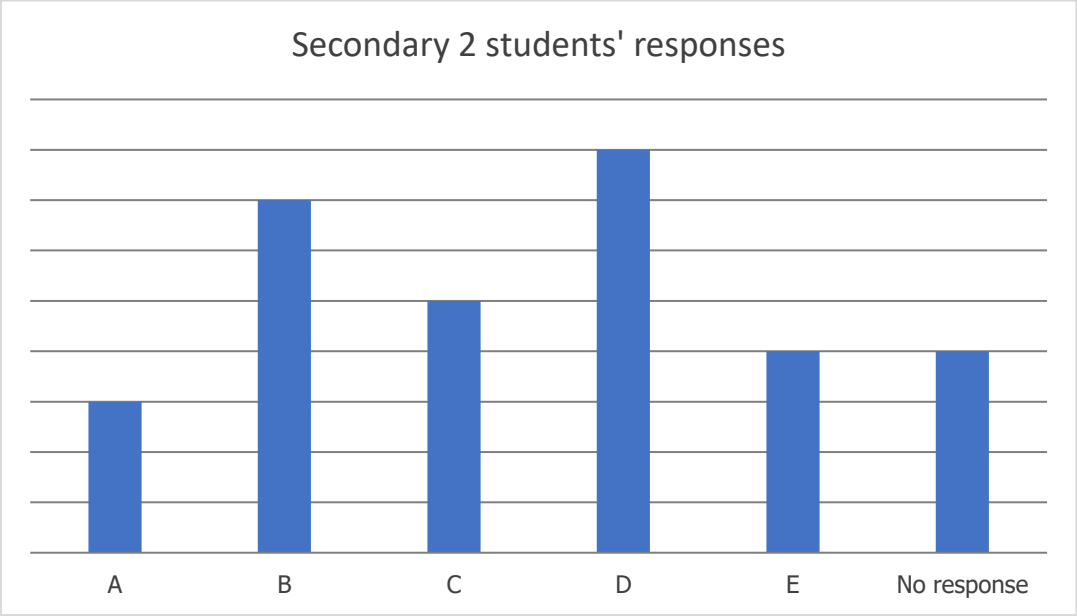
A. 0
B. 6
C. 7
D. 10
E. None of the above

Section B (Correct answer – 4 points| Incorrect or No answer – 0 points)

When an answer is a 1-digit number, shade "0" for the tens, hundreds and thousands place.
Example: if the answer is 7, then shade 0007
When an answer is a 2-digit number, shade "0" for the hundreds and thousands place.
Example: if the answer is 23, then shade 0023
When an answer is a 3-digit number, shade "0" for the thousands place.
Example: if the answer is 785, then shade 0785
When an answer is a 4-digit number, shade as it is.
Example: if the answer is 4196, then shade 4196

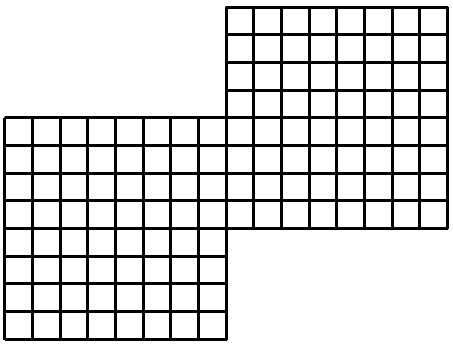
Question 16

The following bar chart shows the responses made by all Secondary 2 students from Integrity Secondary School in a multiple-choice question. All the horizontal lines are equally spaced. The correct answer scores 2 points, 0 points for no response and –1 point for the wrong answer. The total points scored by all the students is –240. If the correct answer is B, how many students solved the question correctly?



Question 17

How many squares are there in the figure below?



Question 18

Chloe and her cousin left home for the cinema at 11.40 a.m. Chloe walked at 100 metres/minute while her cousin walked at 60 m/min. Ten minutes after arriving at the cinema, Chloe realised she had forgot her wallet and so she walked back home. On her way home, Chloe met her cousin who was still on the way to cinema, at 12.10 p.m. How far away was Chloe's home from the cinema, in metres?

Question 19

Find the sum of all positive integers n so that $n^2 + 11n + 52$ is a perfect square.

Question 20

The six-digit number 7A34B4 is divisible by 36. If B is a prime number, find the value of A.

Question 21

The operator $\wedge$ acts on two integers to give the following outcomes:

$$4\wedge 8 = 2412$$

$$5\wedge 9 = 2812$$

$$1\wedge 7 = 1618$$

$$2\wedge 3 = 103$$

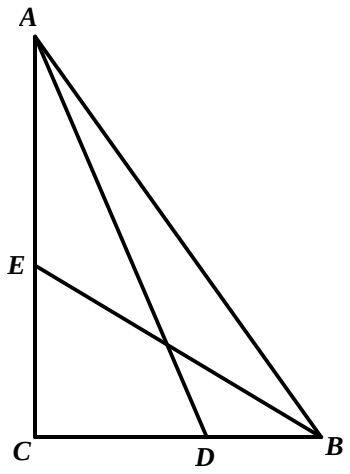
What is the value of $7\wedge 8$?

Question 22

An election for the mayoralty post consists of 5 candidates: A, B, C, D, and E. There are 1400 eligible voters. In the middle of the manual count, A got 300 votes, B got 154 votes, C got 194 votes, D got 281 votes, and E got 266 votes. There were also 9 invalid votes. What is the least number of votes that A needs to get in order to ensure that he wins the mayoralty post?

Question 23

In the diagram, ABC is a right-angled triangle with $\angle C = 90°$. Points D and E are on the segments BC and AC such that $BD:DC = 2:3$, $AE:EC = 3:2$, $AD = 4$, and $BE = \sqrt{10}$. Find the value of $57 \times AB^2$.



Question 24

Two-way roads are constructed between Town A, Town B and Town C. It is given that the number of direct roads between any two of the towns is at least three and at most ten. One can go from Town A to Town C directly or via Town B in a total of 33 different ways. One can go from Town B to Town C directly or via Town A in a total of 23 different ways. In how many ways can one go from Town B to Town A?

Question 25

In the following cryptarithm, all the different letters stand for different digits.

$$\begin{array}{cccccc} & & & W & I & N \\ + & S & A & S & M & O \\ \hline & M & E & D & A & L \\ \hline \end{array}$$

If N = 4 and M = 6, find the value of the sum M + E + D + A + L.

END OF PAPER

Solutions to SASMO 2019 Secondary 2 (Grade 8)

Question 1

$6! \div 5 = 6 \times 5 \times 4 \times 3 \times 2 \times 1 \div 5 = 144$

$6! \div 5 - 4 - 3 - 2 - 1 = 144 - 10 = 134$

$9 + (8 + 7) \times (6! \div 5 - 4 - 3 - 2 - 1) = 9 + 15 \times 134 = \mathbf{2019}$

Answer: **(B)**

Question 2

A. Wrong. $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

B. Wrong. $-10^2 = -100$

C. Wrong. $x + 3y = \sqrt{x^2 + 6xy + 9y^2}$

D. Wrong. $\frac{x}{x} + \frac{x}{y} = 1 + \frac{x}{y} = \frac{y+x}{y}$

Thus, the answer is **Option E**.

Answer: **(E)**

Question 3

There are only 3 pairs of such numbers: (3, 18), (6, 15) and (9, 12). LCM(3, 18) = 18, LCM(6, 15) = 30 and LCM(9, 12) = 36. The answer is **7**.

Answer: **(D)**

Question 4

If a number is divisible by 36 ($= 9 \times 4$), then it is also divisible by 9 and 4, since 9 and 4 are relatively prime.

Using the divisibility test for 4, the last two digits $8D$ is also divisible by 4.

Since D is a single digit, $D = 0, 4 \text{ and } 8.$

Using the divisibility test for 9, $5 + D + 6 + 8 + D = 19 + 2D$ is also divisible by 9.

Since $19 + 2 \times 0$ and $19 + 2 \times 8$ are not divisible by 9, then D = **4**.

Answer: **(C)**

Question 5

It is obvious that $X + Y > X - Y$ and $X + Y > X \times Y$. It is enough to just compare $X + Y$ and $X \div Y$.

$X + Y = 0.0000925$

$\boldsymbol{X \div Y} = \frac{125}{10000000} \div \frac{8}{100000} = \frac{125}{800} = 0.15625$ has the greatest value.

Answer: **(D)**

Question 6

Let the original price of an ice cream cone be P. The amount of money that used to buy the same 60 ice cream cones is $60P$.

The new price $P_{new} = 1.2P$ and $\frac{60P}{P_{new}} = \frac{60P}{1.2P} = \mathbf{50}$ ice cream cones can now be bought using the same amount of money which is $60P$.

Answer: **(A)**

Question 7

If $x \geq y$ and $z > y$, then it does not always mean that $x \geq z$. For example, it is true that $3.14159 \ldots \geq 3$ and $2019 > 3$, but it is clear $3.14159 \ldots < 2019.$

Therefore, the transition step from **(3) to (4)** is wrong.

Answer: **(C)**

Question 8

A. True. $(2 + 4 + \cdots + 200) = 10100 > (1 + 3 + \cdots + 199) = 10000$

B. True. The digit "0" appears 31 times where as other digits appear at least 40 times.

C. True. $\frac{(2+4+\cdots+200)}{100} = \frac{10100}{100} = 101 > \frac{(1+3+\cdots+199)}{100} = \frac{10000}{100} = 100$

D. **NOT true**. $10100 - 10000 = 100 \neq 200$

E. True. $\frac{1+2+3+\cdots+200}{200} = \frac{201}{2}$

Answer: **(D)**

Question 9

Option A cannot be formed since the faces with diagonal and middle lines are swapped when the net is folded.

Option C cannot be formed since the 3 shown faces in the option do not share a common vertex when the net is folded.

Option D cannot be formed since the faces with white and grey squares must be opposite to each other when the net is folded.

Option E cannot be formed since the faces with diagonals must have different diagonals.

Option B can be formed.

Answer: **(B)**

Question 10

Notice that $2 \times 5 = 10$, $2^2 \times 5^2 = 100$, etc., i.e. a zero at the end is produced by the product of a factor of 2 and a factor of 5.

$2^{11} \times 3^7 \times 5^5 \times 7^3 \times 11^2$ will contain five 5s and more than five 2s among its factors.

There are at most 5 pairs of factor 2 and factor 5, i.e. there are **5** consecutive zeros at the end of $2^{11} \times 3^7 \times 5^5 \times 7^3 \times 11^2$!

Answer: **(D)**

Question 11

Let a be the speed of the first cyclist and b be the speed of the second cyclist. The distance travelled by the first cyclist in 3 hours is $3a$ $(Speed \times Time = Distance)$. The distance travelled by the second cyclist in 7 hours is $7b$. The total distance they travel is

$$3a + 7b = 150 \quad (1)$$

We also know that

$$a + b = 34 \quad (2)$$

Solving equations (1) and (2), we get $b = 12$, $a = 22$ and $a - b = 22 - 12 = \mathbf{10}$ **km/h**.

Answer: **(C)**

Question 12

$$1 + 2 + 3 + \cdots + 100 = \frac{100 \times 101}{2} = 5050$$

$$5041 = 71^2 < 1 + 2 + 3 + \cdots + 100 < 72^2 = 5184$$

Therefore, $\sqrt{1 + 2 + 3 + \cdots + 100} = 71.abc$ ... and $\left[\sqrt{1 + 2 + 3 + \cdots + 100}\right] = \mathbf{71}$.

Answer: **(C)**

Question 13

We can observe that if we double the numerator and subtract 1 from the answer, we will get the value of the denominator; i.e., $\frac{60}{119} = \frac{60}{2 \times 60 - 1}$.

We now take the reciprocal of each fraction, the new fractions are

$$\frac{119}{60}, \frac{127}{64}, \frac{123}{62}, \frac{125}{63}, \frac{121}{61} \; or \; \frac{120-1}{60}, \frac{128-1}{64}, \frac{124-1}{62}, \frac{126-1}{63}, \frac{122-1}{61} \; or$$

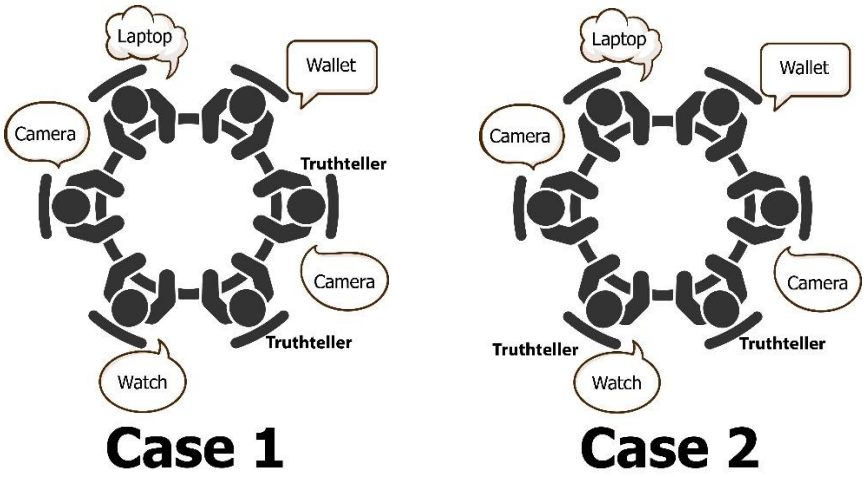
$$\left(2 - \frac{1}{60}\right), \left(2 - \frac{1}{64}\right), \left(2 - \frac{1}{62}\right), \left(2 - \frac{1}{63}\right), \left(2 - \frac{1}{61}\right)$$

Of which, the smallest value is $\left(2 - \frac{1}{60}\right)$. Hence, its reciprocal yields the largest fraction in the question, which is $\mathbf{\frac{60}{119}}$.

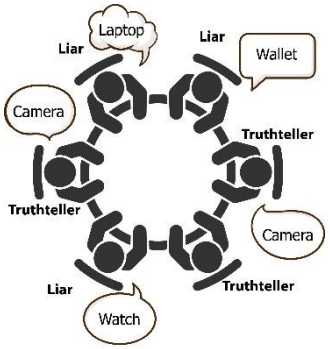
Answer: **(A)**

Question 14

There are only 2 possible ways (shown below) for 2 truth-tellers to sit next to each other.



Let's assume that Case 1 is correct. Then there are 3 truth-tellers and 3 liars as shown on the right. Two liars who responded "Laptop" and "Wallet" are sitting next to each other. The remaining two followers, who responded, "Camera" and "Watch", are also sitting next to each other. This contradicts the question statement. Therefore, Case 1 is incorrect.

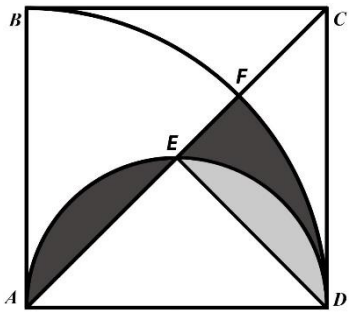


From Case 2, we derive that **Watch** is inside the gift box.

Answer: **(D)**

Question 15

Let E and F be the intersection points of the semicircle and quadrant with the diagonal AC, respectively. The areas of semicircles segments AE and ED are equal. Hence, we just need to find the area of the shaded figure EFD in the quadrant.



The area of the sector FAD is half the area of the quadrant or $\frac{20^2\pi}{8} = 50\pi$.

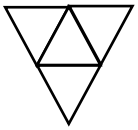
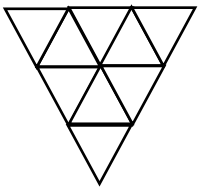
The area of the triangle EAD is quarter of the area of the square or $\frac{20^2}{4} = 100$.

(Area of the shaded figure EFD) = (Area of the sector FAD) – (Area of the triangle EAD) = $(\mathbf{50\pi - 100})$ cm^2.

Answer: **(D)**

Question 16

The following are the numbers of triangles:

Type of triangle	1-part	4-part	9-part	16-part
Example				
Number	30	18	8	3

The total number of triangles is $30 + 18 + 8 + 3 = \mathbf{59}$.

Answer: **59**

Question 17

Let the time, in minutes, Lou spent in the gym last week Wednesday be x.

The average time of the first 3 days is $\frac{21+14+x}{3}$. The average time of the last 4 days is $\frac{1}{0.72} \times \frac{21+14+x}{3}$. Hence the total time spent in the gym in the last 4 days is $4 \times \frac{1}{0.72} \times \frac{21+14+x}{3}$.

Given the total time spent in the gym last week is

$$21 + 14 + x + 4 \times \frac{1}{0.72} \times \frac{21 + 14 + x}{3} = 183$$

Solving the equation, we get $x = 29.1688311688$. Rounding off x to the nearest integer, we get **29**.

Answer: **29**

Question 18

Since $2018! = 2018 \times 2017 \times \ldots \times 2 \times 1$, it is clear that $2018!$ is divisible by $2019 = 673 \times 3$ (673 and 3 are relatively prime). Thus, when $2018! + 13$ is divided by 2019, the remainder is $\mathbf{13}$.

Answer: **13**

Question 19

The pattern is as follows:

$$a \otimes b = (b-2)\left(\frac{a+b}{2}\right),$$

where we don't multiply the brackets, but write them next to each other.

$$2 \otimes 6 = (6-2)\left(\frac{2+6}{2}\right) = 44$$

$$4 \otimes 10 = (10-2)\left(\frac{4+10}{2}\right) = 87$$

$$8 \otimes 18 = (18-2)\left(\frac{8+18}{2}\right) = 1613$$

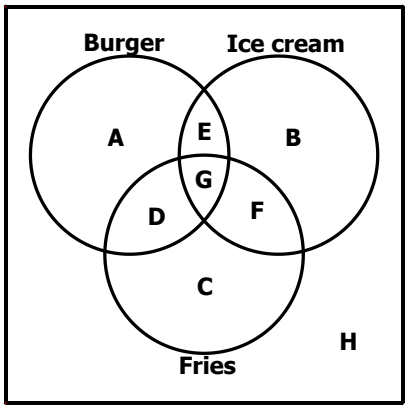
$$16 \otimes 22 = (22-2)\left(\frac{16+22}{2}\right) = 2019$$

Therefore, $7 \otimes 9 = (9-2)\left(\frac{7+9}{2}\right) =$ **78**

Answer: **78**

Question 20

Draw a Venn Diagram as shown in the diagram on the right.



A, B and C are the numbers of students who received only Burger, Ice Cream and Fries, respectively. H is the number of students who didn't get any of the three types of food.

31 students received all the three types of food or $\mathrm{G} = 31$.

Fifteen students received burger and ice cream but not fries or $E = 15$.

134 students received a burger or

$$\mathrm{A} + \mathrm{D} + \mathrm{G} + \mathrm{E} = 134$$

62 received an ice cream or

$$\mathrm{B} + \mathrm{F} + \mathrm{G} + \mathrm{E} = 62$$

Since $\mathrm{G} = 31$ and $E = 15$, then $\mathrm{B} + \mathrm{F} = 62 - 31 - 15 = 16$

There are 200 students in this camp or

$$200 = \mathrm{A} + \mathrm{B} + \mathrm{C} + \mathrm{D} + \mathrm{E} + \mathrm{F} + \mathrm{G} + \mathrm{H} = (\mathrm{A} + \mathrm{D} + \mathrm{G} + \mathrm{E}) + (\mathrm{B} + \mathrm{F}) + \mathrm{C} + \mathrm{H}$$
$$= 134 + 16 + \mathrm{C} + \mathrm{H}$$

Hence $\mathrm{C} + \mathrm{H} = 50$

For H to be the largest possible, C must be the smallest possible which is 0. Therefore, the largest possible number of students who didn't get any of the three types of food is $\mathrm{H} = \mathbf{50}$.

Answer: **50**

Question 21

The prime factorisation of 360 is $2^3 \times 3^2 \times 5$. If X is a 6-digit number, then the smallest product of its digits must be $1 \times 2 \times 3 \times 4 \times 5 \times 6 = 720$. Hence X must be a 5-digit number so that the value of $\frac{X}{10}$ is the greatest possible.

The greatest possible value of X is 95421 and the product of its digits is $9 \times 5 \times 4 \times 2 \times 1 = 360$. By Divisibility Rule of 3, 95421 is divisible by 3, so it is a composite number. Rounding off $\frac{X}{10}$ to the nearest integer, we get **9542**.

Answer: **9542**

Question 22

We note that

$$\frac{n-1}{n!} = \frac{n}{n!} - \frac{1}{n!} = \frac{1}{(n-1)!} - \frac{1}{n!}$$

$$\frac{2}{1 \times 2 \times 3} + \frac{3}{1 \times 2 \times 3 \times 4} + \cdots + \frac{6}{1 \times 2 \times 3 \times \ldots \times 7} =$$
$$= \frac{3-1}{1 \times 2 \times 3} + \frac{4-1}{1 \times 2 \times 3 \times 4} + \cdots + \frac{7-1}{1 \times 2 \times 3 \times \ldots \times 7}$$
$$= \left(\frac{1}{2!} - \frac{1}{3!}\right) + \left(\frac{1}{3!} - \frac{1}{4!}\right) + \cdots + \left(\frac{1}{6!} - \frac{1}{7!}\right) = \frac{1}{2!} - \frac{1}{7!}$$

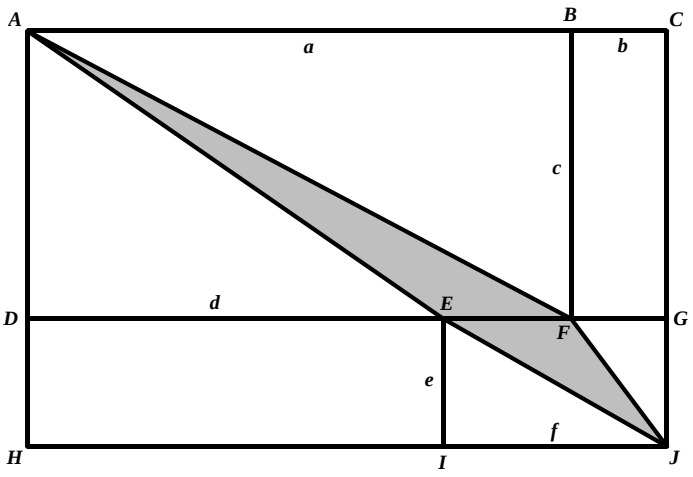
Hence

$$15120 \times \left(\frac{2}{1 \times 2 \times 3} + \frac{3}{1 \times 2 \times 3 \times 4} + \cdots + \frac{6}{1 \times 2 \times 3 \times \ldots \times 7}\right) = 15120 \times \left(\frac{1}{2!} - \frac{1}{7!}\right)$$
$$= \mathbf{7557}$$

Answer: ***7557***

Question 23

Label the vertices and segments as shown in the diagram.



It is given that

$$ac = 160$$

$$bc = 40$$

$$de = 30$$

$$fe = 10$$

From the equations above, we derive that

$160 + 40 = 200 = ac + bc = (a + b)c$ (1)

$30 + 10 = 40 = de + fe = (d + f)e$ (2)

Dividing equation (1) by (2), we get $5 = \frac{c}{e}$ since $a + b = d + f$.

$$Area(AEF) = Area(ABFD) - Area(ABF) - Area(ADE) = 160 - \frac{ac}{2} - \frac{cd}{2}$$
$$= 160 - 80 - \frac{cd}{2} = 80 - \frac{cd}{2}$$

$$Area(JEF) = Area(EGIJ) - Area(EIJ) - Area(FGJ) = 10 - \frac{fe}{2} - \frac{be}{2} = 10 - 5 - \frac{be}{2}$$
$$= 5 - \frac{be}{2}$$

$$Area(AEF) + Area(JEF) = 80 - \frac{cd}{2} + 5 - \frac{be}{2} = 85 - \left(\frac{cd}{2} + \frac{be}{2}\right)$$

From $5 = \frac{c}{e}$, we get that $c = 5e$ and $e = \frac{c}{5}$. By substituting c and e, we get

$$Area(AEF) + Area(JEF) = 85 - \left(\frac{cd}{2} + \frac{be}{2}\right) = 85 - \left(\frac{5de}{2} + \frac{bc}{10}\right) = 85 - \left(\frac{5 \times 30}{2} + \frac{40}{10}\right)$$
$$= 85 - 79 = \mathbf{6}$$

Answer: **6 cm²**

Question 24

From the thousands place, it is obvious that $D = G + 1$.

In the hundreds place addition, the ones digit of $D + O$ is G or the ones digit of $D + O + 1$ is G (if there is a carryover from the tens place addition). Hence, the value of O can only be 8 or 9.

Case 1: $O = 9$

In the tens place addition, there is no carryover of 1 from ones place addition otherwise $D = 9 = O$ which is impossible. Therefore, $D = 8$ $(O + O = 18)$ and $G = D - 1 = 7$ which implies that there is a carryover of 1 from ones place addition. We got the contradiction, so Case 1 is impossible.

Case 2: $O = 8$

In the tens place addition, $D = 6\ or\ 7$ $(O + O = 16\ or\ O + O + 1 = 17)$ and $G = 5\ or\ 6$. Hence, $D + G > 10$ and there is a carryover of 1 from ones place addition. We get $D = 7$ $(O + O + 1 = 17)$, $G = 6$ and $A = 3$.

The sum O + D + A + G is equal to $8 + 7 + 3 + 6 = \mathbf{24}$.

Answer: **24**

Question 25

$\overline{cccc} = 1000c + 100c + 10c + c = 1111c$

$\overline{bb} = 10b + b = 11b$

$\overline{aa} = 10a + a = 11a$

Rewrite the equation

$$(11a)^2 + 11b = 1111c$$

Then divide by 11 both sides of the equation

$$11a^2 + b = 101c$$

If $c = 1$, then $b = 2$ and $a = 3$

If $c = 2,$ then there is no integer solution for $a.$

and so on…

By considering all possible value of c, we get the following solutions:

$c = 1$, $b = 2$ and $a = 3$

$c = 4$, $b = 8$ and $a = 6$

$c = 7$, $b = 3$ and $a = 8$

The largest possible value of $a + b + c$ is $7 + 3 + 8 = 4 + 8 + 6 = \mathbf{18}$.

Answer: **18**

Solutions to SASMO 2020 Secondary 2 (Grade 8)

Question 1

$2020 \times 20 - 20 \times 20 + 2020 = 20(2020 - 20) + 2020 = 20 \times 2000 + 2020 = 40000 + 2020 =$ **42020**

Answer: **(B)**

Question 2

$3^1 \times 3^3 \times 3^5 \times 3^7 \times 3^9 \times 3^{11} \times 3^{13} \times 3^{15} \times 3^{17} \times \left(\frac{1}{3}\right)^{75} = 3^{1+3+5+7+\cdots+15+17} \times \left(\frac{1}{3}\right)^{75} = 3^{81} \times \left(\frac{1}{3}\right)^{75} = 3^6 =$**729**

Answer: **(D)**

Question 3

$ax - 2by + 3c = a(3a - 4b^2) - 2b(4b - 3c^2) + 3c = 3a^2 - 4ab^2 - 8b^2 - (-6bc^2) + 3c = 3a^2 - 4ab^2 - 8b^2 + 6bc^2 + 3c$

None of the options in the question correspond to the above expression. Note that in this question, the value of z is redundant information.

Answer: **(E)**

Question 4

$$\frac{1 + 2 + 3 + \cdots + 120}{110} = \frac{\left(\frac{120 \times 121}{2}\right)}{110} = \frac{12 \times 10 \times 11 \times 11}{2 \times 10 \times 11} = 66 = \frac{12 + N}{2}$$

$N = 66 \times 2 - 12 =$ **120**

Answer: **(E)**

Question 5

The length of each side of the large rectangle is represented by A, B, C, D, E and F respectively, as shown in the diagram below.

	A	*B*	C
D	**W**	**1**	**X**
E	$\frac{1}{6}$	**Y**	$\frac{1}{2}$
F	$\frac{1}{3}$	$\frac{2}{3}$	**Z**

Observe that AE = $\frac{1}{6}$ and AF = $\frac{1}{3}$, or F = 2E. Hence, area of Y = BE = $\frac{1}{2}\times$ BF = $\frac{1}{2}\times\frac{2}{3}=\frac{1}{3}$ and area of Z = CF = $2\ \times$ CE = $2\times\frac{1}{2}=1$.

Observe that AF = $\frac{1}{3}$, and BF = $\frac{2}{3}$, or B = 2A. Hence, area of W = AD = $\frac{1}{2}\times$ BD = $\frac{1}{2}\times 1=\frac{1}{2}$.

Finally, observe that AE = $\frac{1}{6}$ and CE = $\frac{1}{2}$, or C = 3A. Hence, area of X = CD = $3\times$ AD = $3\times\frac{1}{2}=\frac{3}{2}$.

Sum of area of remaining 4 rectangles = $\frac{1}{3}+1+\frac{1}{2}+\frac{3}{2}=\mathbf{\frac{10}{3}}$

Answer: **(A)**

Question 6

$$8\Delta x=(8-1)(8-2)\ldots(8-x)=7\times 6\times\ldots\times(8-x)=840$$

Note that $840=7\times 120=7\times 6\times 20=7\times 6\times 5\times 4$ so $x=8-4=4$

$x\Delta(x-3)=4\Delta(4-3)=4-1=$ **3**

Answer: **(E)**

Question 7

Let $n = 2021$

Then $\frac{2022^3-2\times2022^2-2020}{2021^3+2\times2021^2-2023} = \frac{(n+1)^3-2(n+1)^2-n+1}{n^3+2n^2-n-2} = \frac{n^3+3n^2+3n+1-2n^2-4n-2-n+1}{(n-1)(n^2+3n+2)} =$
$\frac{n^3+n^2-2n}{(n-1)(n+1)(n+2)} = \frac{n(n-1)(n+2)}{(n-1)(n+1)(n+2)} = \frac{n}{n+1} = \frac{2021}{2022}$

$a + b =$ **4043**

Answer: **(C)**

Question 8

If $x < -4$, then $|x + 3| = -x - 3$

$\left|1 - \sqrt{(x+3)^2}\right| = |1 - |x + 3|| = |1 - (-x - 3)| = |x + 4| = -\boldsymbol{x} - \mathbf{4}$ (since $x < -4$)

Answer: **(D)**

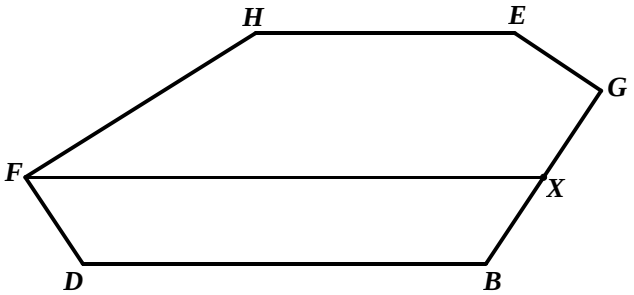
Question 9

The sum of the interior angles in a n-sided figure (or polygon) is $(n - 2) \times 180°$.

Draw line FX, which is parallel to lines HE and DB.

Then $\angle DFX = 90° - (180 - 138)° = 48°$ This means that $\angle D = 180° - \angle DFX = 180° - 48° = 132°$ and $\angle B = 132° + 4 = 136°$

Thus, $\angle E = (6 - 2) \times 180° - \angle G - \angle B - \angle D - \angle F - \angle H = 720° - 90° - 136° - 132° - 90° - 138° = \mathbf{134}°$



Answer: **(A)**

Question 10

Since $2020 = 2^2 \times 5 \times 101$, any fraction which numerator is a multiple of 2, 5 OR 101 will not be in its simplest form.

From 1 to 200, only 1 multiple of 101, or 101, exists. Hence there is only 1 fraction whose numerator is a multiple of 101.

From 1 to 200, there exists $\frac{200}{2} = 100$ multiples of 2, $\frac{200}{5} = 40$ multiples of 5, and $\frac{200}{10} = 20$ multiples of 10. Number of fractions which numerator is a multiple of 2 OR 5 = $100 + 40 - 20 = 120$

Number of fractions not in their simplest form = $120 + 1 = 121$, this means that $200 - 121 =$ **79** fractions are in their simplest form.

Answer: **(C)**

Question 11

The implicit condition of the question is that any 5-digit numbers using all digits 1,2,3,4 and 5 (this will make a total of 5! = 120 such numbers without restrictions) can be written, so long as the digit 2 comes before digit 4.

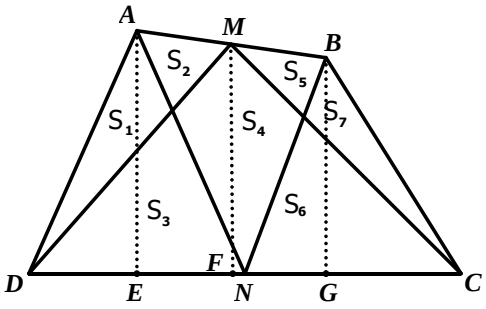
Notice that for every 1 number which digit 2 comes before digit 4, there will be one "counterpart" number which digit 4 comes before digit 2, while retaining the same position of its odd digits. For example, for 32145, its counterpart would be 34125.

This means that the number of numbers whose digit 2 comes before digit 4 = number of numbers whose digit 4 comes before digit 2, so Boris can write $\frac{120}{2} =$**60** numbers.

Answer: **(C)**

Question 12

From A, M, and B, draw 3 lines perpendicular to DC and label them as $AE = h_a, MF = h_m,$ and $BG = h_b$ respectively.



Since M is the midpoint of AB, MF is a midline. We have $h_m = \frac{1}{2}(h_a + h_b)$ or $h_a + h_b = 2h_m$. Also,

$Area(ADN) + Area(BNC) = \frac{1}{2} \times h_a \times DN + \frac{1}{2} h_b \times CN = \frac{1}{2} \times h_a \times DN + \frac{1}{2} h_b \times DN$ (since $CN = DN$)

$$= \frac{1}{2} \times DN \times (h_a + h_b)$$

$$Area(DMC) = \frac{1}{2} \times h_m \times DC = \frac{1}{2} \times \frac{1}{2} \times (h_a + h_b) \times (DN + NC)$$
$$= \frac{1}{2} \times \frac{1}{2} \times (h_a + h_b) \times (2DN) = \frac{1}{2} \times (h_a + h_b) \times DN$$

Hence $Area(ADN) + Area(BNC) = Area(DMC)$ or $S_1 + S_3 + S_6 + S_7 = S_3 + S_4 + S_6$ or $\boldsymbol{S_1 + S_7 = S_4}$.

Answer: **(B)**

Question 13

We will attempt to find a pattern when 2^n is divided by 7 (because there are 7 days in a week), where n is a positive integer.

When $n = 1$, the remainder is 2.

When $n = 2$, the remainder is 4.

When $n = 3$, the remainder is 1.

When $n = 4$, the remainder is 2.

When $n = 5$, the remainder is 4.

When $n = 6$, the remainder is 1.

When $n = 7$, the remainder is 2.

This shows that 2^n leaves a remainder of 2 when divided by 7 if n leaves a remainder of 1 when divided by 3. Since 2020 leaves a remainder of 1 when divided by 3, 2^{2020} leaves a remainder of 2 when divided by 7. The question is now equivalent to finding which day is 2 days after Tuesday, which will be **Thursday.**

Answer: **(B)**

Question 14

Assume Ellie and Audrey are from Westwood Secondary School, then Tom, Chris and Violet are from Greenwich Secondary school. This contradicts the fact that Tom and Chris are from different schools. Hence, Ellie and Audrey are from Greenwich Secondary School.

Similarly, assume Violet and Chris are Secondary 3 students, then Audrey, Tom and Ellie are Secondary 1 students. This contradicts the fact that Audrey and Tom study at different levels. Hence, Violet and Chris are Secondary 1 students.

To find the Secondary 3 student from Westwood Secondary School, we can eliminate Violet and Chris who are Secondary 1 students, and Ellie and Audrey who are from Greenwich Secondary school. This leaves **Tom** as the only student who fulfils the question's criteria.

Answer: **(D)**

Question 15

Note that 5!, 6!, 7!, 8! and 9!, have one consecutive digit '0' at the end since each of them contain exactly one '5' in the product.

Next, 10! to 11! have two consecutive digit '0's at the end since each of them contain exactly two '5's in the product.

We continue listing and notice that 20! to 21! have 4 consecutive digit '0's at the end, but 25! to 29! have 6 consecutive digit '0's at the end. Thus, it is impossible to have 5 zeroes at the end of $n!$.

Similarly, we obtain that it is impossible to have 11, 17, 23, 29, 30 or 36 zeroes at the end of $n!$.

Thus, there are **7 integers** from 1 to 40 which cannot be the number of consecutive digit "0"s at the end of $n!$.

Answer: **(C)**

Question 16

Let each space of the bar graph be x students.

Number of students who answered the question correctly = number of students who chose B = $7x$. Hence, a total of $14x$ points were scored from them.

Number of students who answered the question wrongly = number of students who chose A, C, D or E = $3x + 5x + 8x + 4x = 20x$. A total of $-20x$ points were scored from them.

Total number of points =$14x - 20x = -6x = -240$, or $x = 40$.

Number of students who answered the question correctly = $7x = \mathbf{280}$

Answer: **280**

Question 17

Notice the figure is made of 2 overlapping 8 x 8 squares.

We first attempt to calculate the number of possible squares that can be formed in each 8 x 8 square:

Number of 1 x 1 squares = 8^2

Number of 2 x 2 squares = 7^2

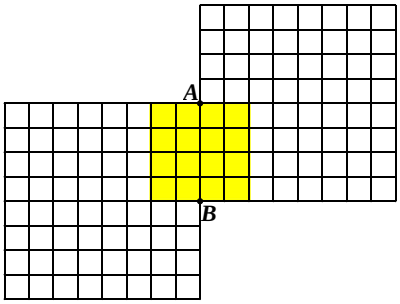
Number of 3 x 3 squares = 6^2

…

Number of 8 x 8 squares = 1^2

Hence total number of squares in two 8 x 8 squares = $2 \times (1^2 + 2^2 + 3^2 + \cdots + 8^2) = 408$

Finally, notice that there is an overlapping region (shaded in yellow in the diagram below) between the 2 squares, which has a width of 4 small squares. From this overlapping region, all squares formed must pass though the line AB. Hence, number of 2 x 2 squares = 3, number of 3 x 3 squares = 4 and number of 4 x 4 squares = 3.



Total number of squares = 408 + 3 + 4 + 3 = **418**

Answer: **418**

Question 18

Let the time Chloe took to reach the cinema be x minutes, then distance from home to the cinema is $100x$ metres.

At 12.10 p.m., 30 minutes after 11.40 a.m., her cousin had walked $60 \times 30 = 1800$ metres towards the cinema, while Chloe had walked $100(30 - x - 10) = (2000 - 100x)$ metres towards home. Notice that this was when Chloe and her cousin had met, which means that the total distance Chloe and her cousin covered at 12.10 p.m. is the distance between Chloe's home and the cinema.

$$2000 - 100x + 1800 = 100x$$

$x = 19$ minutes. Hence, the distance is **1900** metres.

Answer: **1900**

Question 19

Let $n^2 + 11n + 52 = x^2$ for some positive integer x.

We complete the square for the equation $n^2 + 11n + 52 = x^2$ to get $\left(n + \frac{11}{2}\right)^2 + \frac{87}{4} = x^2$ or $\frac{(2n+11)^2}{4} + \frac{87}{4} = x^2$

Multiply the above equation by 4, and we can express $m = 2n + 11$ to get $m^2 + 87 = (2x)^2$ or $(2x + m)(2x - m) = 87$. Since $2x + m > 2x - m$, and $2x + m$ and $2x - m$ are positive integer factors of 87,

$2x + m = 29$ and $2x - m = 3$, which gives $m = 13$, or $2x + m = 87$ and $2x - m = 1$, which gives $m = 43$. When $m = 13$, $n = \frac{13-11}{2} = 1$. When $m = 43, n = \frac{43-11}{2} = 16$

Hence, the sum of all such n is 1+16=**17**.

Answer: **17**

Question 20

If 7A34B4 is divisible by 36, it must be divisible by 4. By divisibility rule, B4 must be divisible by 4. Since B is a prime, B = 2.

If 7A34B4 is divisible by 36, it must be divisible by 9. By divisibility rule, 7 + A + 3 + 4 + 2 + 4, or 20 + A must be divisible by 9.

Since A is a one-digit number, A = **7**

Answer: **7**

Question 21

The pattern is as follows: $a\Lambda b = [2(a+b)][3(b-a)]$

where we don't multiply the square brackets, but write these numbers next to each other. E.g. $4\Lambda 8 = [2(8+4)][3(8-4)] = 2412$

This is a very common "trick" in SASMO, and must be known by heart!

$7 \Lambda 8 = [2(7+8)][3(8-7)] = \mathbf{303}$

Answer: **303**

Question 22

In the middle of the manual count, there are $1400 - 300 - 154 - 194 - 281 - 266 - 9 = 196$ votes which are yet to be counted.

To ensure A wins the post, we consider the worst-case scenario, when all remaining votes are either for A, or for the candidate with the 2nd highest vote count, D.

In this scenario, D needs 20 more votes than A (since A is currently leading by 19 votes), from the remaining 196 votes to win, or $\frac{196+20}{2} = 108$ votes.

Hence A needs a minimum of $300 + 196 - 108 + 1 = \mathbf{389}$ votes to win.

Answer: **389**

Question 23

Let the length of AC be $5x$ and the length of BC be $5y$.

By Pythagoras theorem,

$EC^2 + BC^2 = (2x)^2 + (5y)^2 = 4x^2 + 25y^2 = BE^2 = 10$, and

$$AC^2 + CD^2 = (5x)^2 + (3y)^2 = 25x^2 + 9y^2 = AD^2 = 16$$

Solving the simultaneous equations, we get $x^2 = \frac{10}{19}$ and $y^2 = \frac{6}{19}$

One final application of Pythagoras theorem reveals that $AB^2 = AC^2 + BC^2 = (5x)^2 + (5y)^2 = 25(x^2 + y^2) = 25 \times \left(\frac{10}{19} + \frac{6}{19}\right) = 25 \times \frac{16}{19} = \frac{400}{19}$

$57 \times AB^2 = 57 \times \frac{400}{19} = \mathbf{1200}$

Answer: **1200**

Question 24

Let there be

- c roads between Town A and B
- a roads between Town B and C
- b roads between Town C and A

One can go from Town A to Town C directly or via Town B in a total of 33 different ways:

$$b + ca = 33 \quad (1)$$

One can go from Town B to Town C directly or via Town A in a total of 23 different ways:

$$a + bc = 23 \ (2)$$

Subtracting (2) from (1):

$$33 - 23 = 10 = b + ca - a - bc = b - a + c(a - b) = (c - 1)(a - b)$$

Since $10 \geq a, b, c \geq 3$,then $(c - 1) = 2\ or\ 5$, $c = 3\ or\ 6$ and $(a - b) = 5\ or\ 2$.

From (1),

if $c = 3$ and $a - b = 5$, then $33 = b + ca = b + 3 \times (5 + b) = 15 + 4b \Rightarrow b = 4.5$

However, it is not possible to get 4.5 ways.

if $c = 6$ and $a - b = 2$, then $33 = b + ca = b + 6 \times (2 + b) = 12 + 7b \Rightarrow b = 3$ and $a = 5$

From Town B to Town A, there are $c + ab = 6 + 3 \times 5 = \mathbf{21}$ ways.

Answer: **21**

Question 25

S = 5 because S≠M and there can be a carrying over of maximum 1 to S, so S + 1 = 6. A = 9 because A must carry a 1 over to S, so A + 1 ≥ 10. When A = 9, E = 0.

Substitute N = 4, M = 6, S = 5, A = 9 and E = 0 to get the following:

			W	I	4
+	5	9	5	6	O
	6	0	D	9	L

We are now left with 5 numbers which W, I, O, D and L represent, namely 1,2,3,7 and 8. With these numbers, there are only 2 cases left to consider for 4 + O = L.

Case 1: O = 3 and L = 7. But this will leave no possible value for I to satisfy I + 6 = 9 since O = 3. Hence case 1 is impossible.

Case 2: O = 7 and L = 1. Then I = 2 for 1 + I + 6 = 9. Finally, W = 8 and D = 3 for W + 5 = D + 10

M + E + D + A + L = 6 + 0 + 3 + 9 + 1 = **19**

Answer: **19**